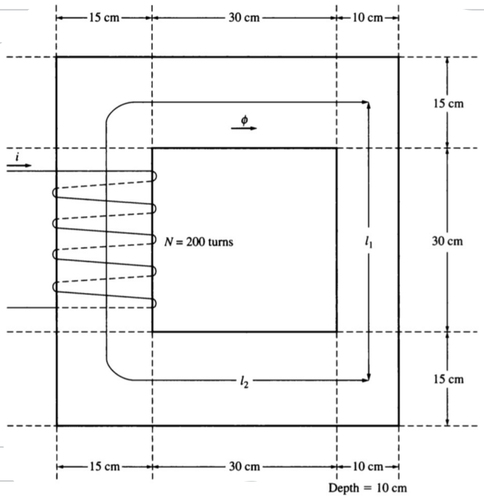
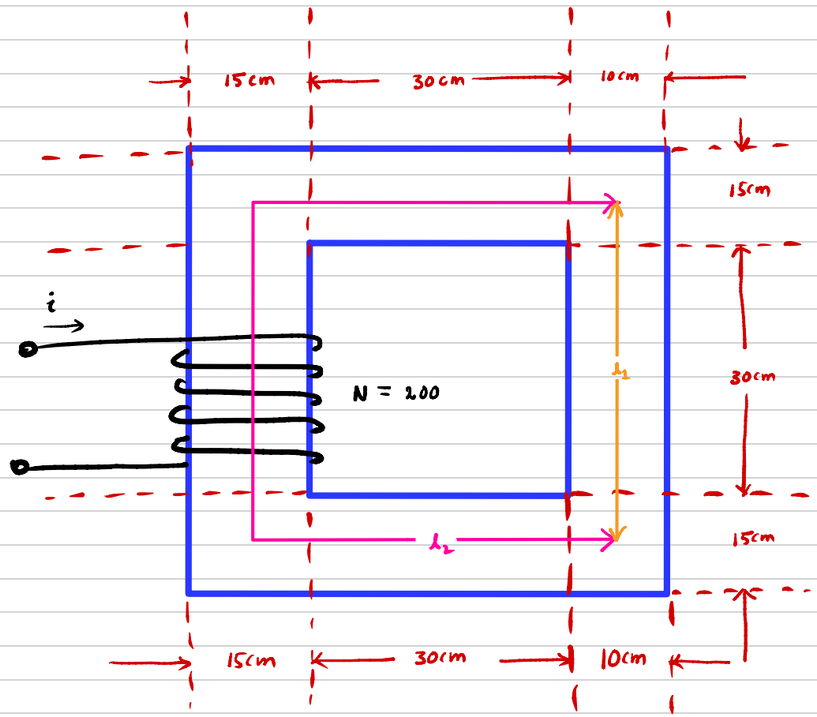
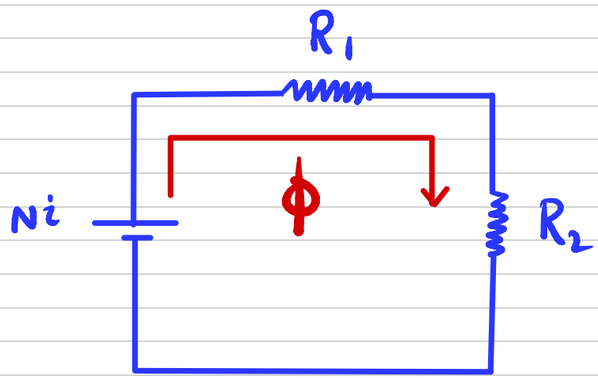
Question 1



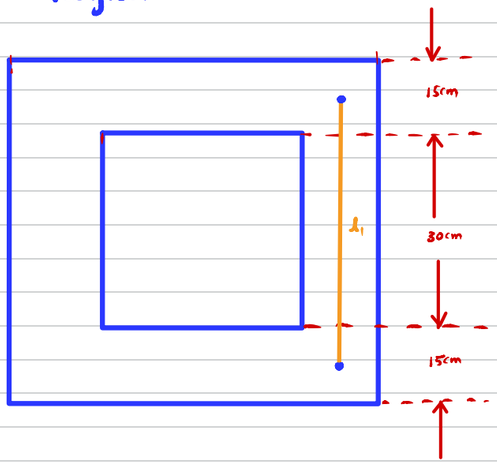


Magnetic Circuit



Reluctance calculation :

Region - 1 :



$$l_1 = 30 + \frac{15}{2} + \frac{15}{2}$$

$$l_1 = 45\text{cm}$$

$$R_1 = \frac{l_1}{\mu_1 A_{c_1}} = \frac{l_1}{\mu_0 \mu_{r_1} A_{c_1}}$$

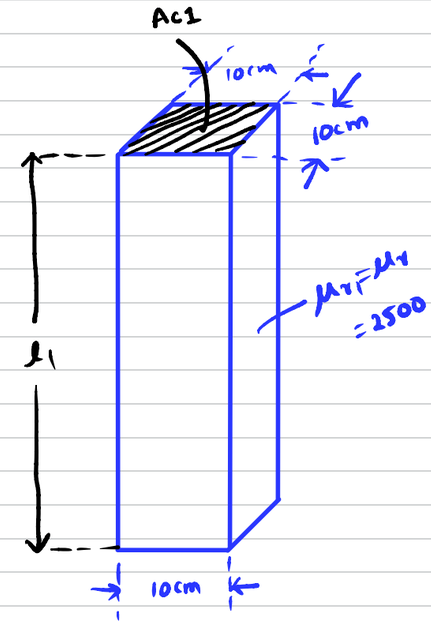
$$A_{c_1} = 10\text{ cm} \times 10\text{cm}$$

$$= 100 \times 10^{-4}\ m^2$$

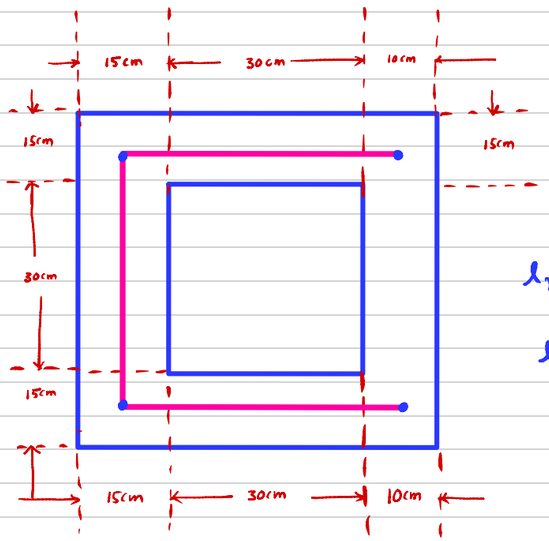
$$A_{c_1} = 0.01\ m^2$$

$$R_1 = \frac{45 \times 10^{-2}}{\mu_0 \times 2500 \times 0.01}$$

$$R_1 = 14323.9449\ \text{At/Wb}$$



Region - 2 :



$$l_2 = 30 \times 3 + \frac{15}{2} \times 4 + \frac{10}{2} \times 2$$

$$l_2 = 130\text{ cm}$$

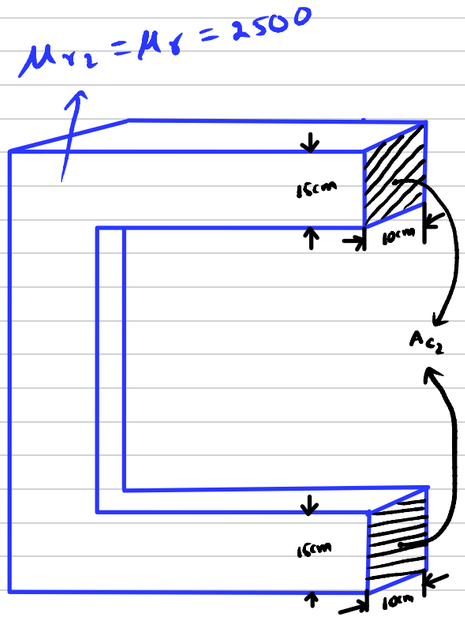
$$R_2 = \frac{l_2}{\mu_2 A_{c_2}} = \frac{l_2}{\mu_0 \mu_{r_2} A_{c_2}}$$

$$A_{c_2} = 15\text{cm} \times 10\text{cm}$$

$$= 0.015\ m^2$$

$$R_2 = \frac{130 \times 10^{-2}}{\mu_0 \times 2500 \times 0.015}$$

$$R_2 = 27586.8568\ \text{At/wb}$$



Flux

$$\sum Ni = \sum \phi R$$

$$Ni = \phi (R_1 + R_2)$$

$$\Rightarrow \quad \phi = \frac{Ni}{R_1 + R_2} = \frac{200 * 1}{14323.9449 + 27586.8568}$$

$$= 0.004772 \text{ Wb}$$

$$\boxed{\phi = 4.772 \text{ mWb}}$$

Question 2

Given

$N = 600$ turns

Diameter of cross-section

$D = 3\,cm$

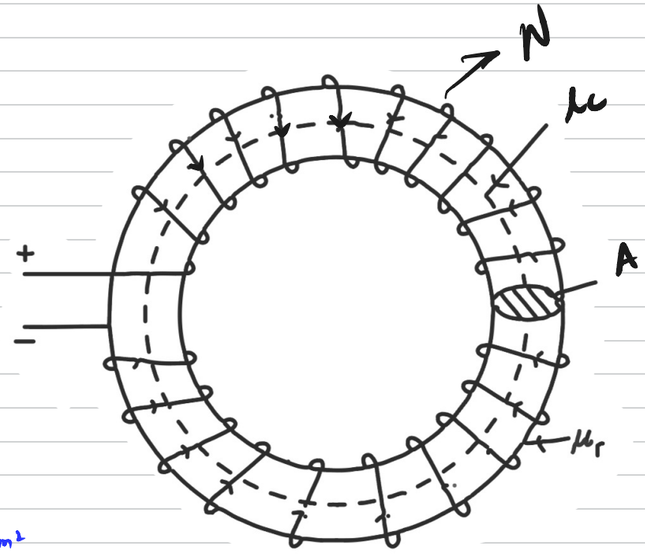
Mean circumference $= 80\,cm = l_c$

$2\pi r = 80\,cm$

$l_c = 80\,cm$

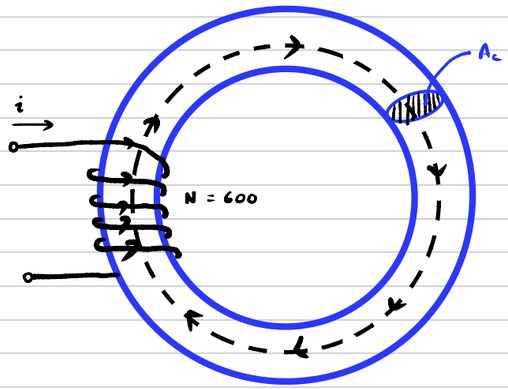
$A_c = \pi r^2 = \pi \left(\frac{3\times10^{-2}}{2}\right)^2$

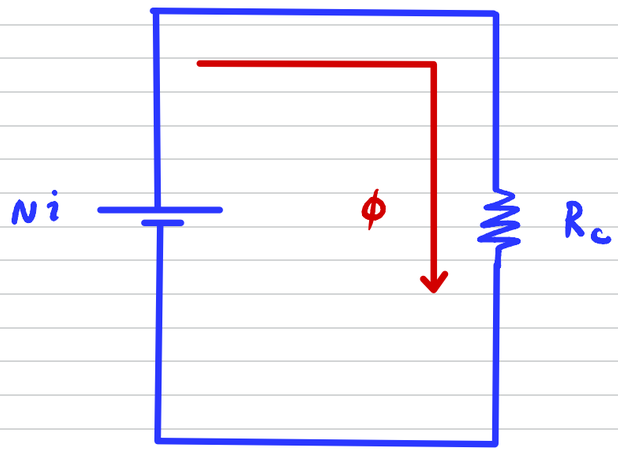
$A_c = 7.0686\times10^{-4}\,m^2$



(a)

Magnetic circuit





$Ni = \phi R_c \Rightarrow \phi = \frac{N i \mu_0 \mu_r A_c}{l_c}$

$i = \frac{\phi R_c}{N} \Rightarrow i = \frac{\phi\, l_c}{N \times \mu_0 \times \mu_r \times A_c}$

(i) $\mu_r = 1$ (air core) ; Given $\phi = 0.5\,mwb$

$i = \frac{0.5\times10^{-3} \times 80\times10^{-2}}{600 \times \mu_0 \times 1 \times 7.0686\times10^{-4}} = 750.527\ Amps$

(ii) $\mu_r = 3000$ (solid cast steel) $\phi = 0.5\,mwb$

$i = \frac{0.5\times10^{-3} \times 80\times10^{-2}}{600 \times \mu_0 \times 3000 \times 7.0686\times10^{-4}} = 0.2502\ Amps$

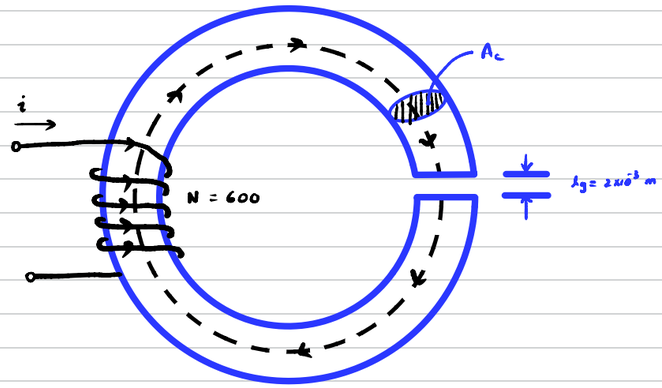
$R_c = \frac{l_c}{\mu A_c} = \frac{80\times10^{-2}}{\mu_0 * \mu_r * A_c}$

(b) air gap length

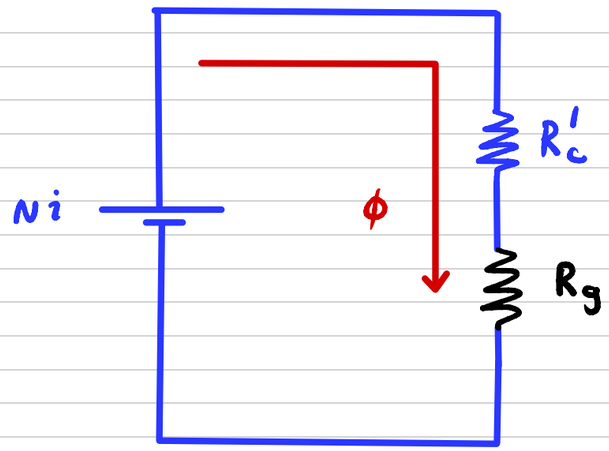
$l_g = 2mm$

$= 2 \times 10^{-3} m$

$i = 0.2502$ Amps (from a.ii.)



Magnetic circuit



Magnetic circuit

$$\phi = \frac{N i}{R_c' + R_g}$$

$$l_c' = 2\pi r - l_g = l_c - l_g$$

$$= 80 \times 10^{-2} - 2 \times 10^{-3}$$

$$= 798 \times 10^{-3}$$

$$R_c' = \frac{l_c'}{\mu_0 \times \mu_r \times A_c} = \frac{798 \times 10^{-3}}{\mu_0 \times 3000 \times 7.0680 \times 10^{-4}}$$

$$= 299460.3872$$

$$R_g = \frac{l_g}{\mu_0 \times 1 \times A_c} = \frac{2 \times 10^{-3}}{\mu_0 \times 1 \times 7.0680 \times 10^{-4}}$$

$$= 2251581.859$$

$$\phi_g = \phi = \frac{N i}{R_c' + R_g} = \frac{600 \times 0.2502}{299460.3872 + 2251581.859}$$

$$= 58.841 \mu wb$$

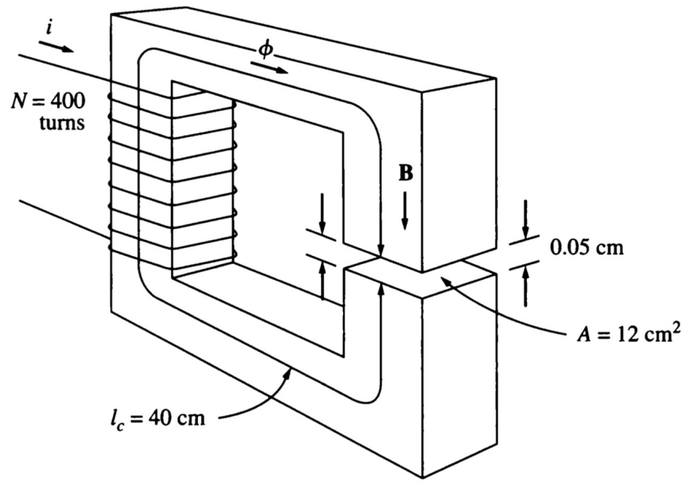
(c) $\phi = 0.5$ mwb

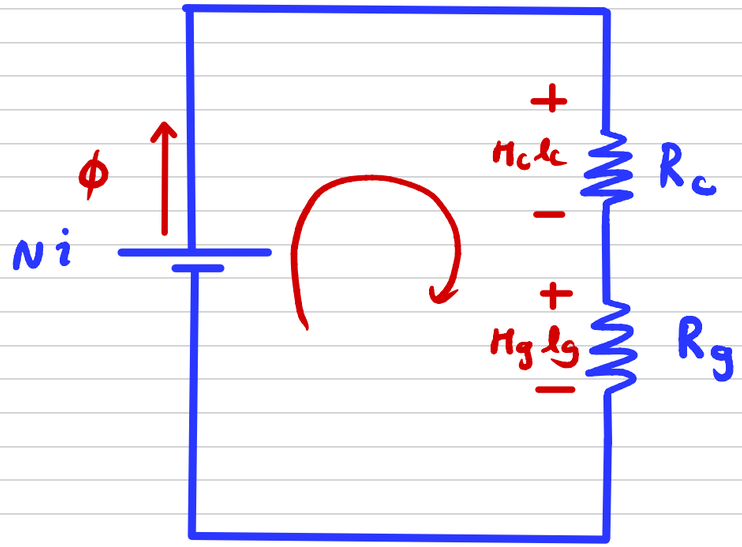
$$i = \frac{\phi (R_c' + R_g)}{N}$$

$$= \frac{0.5 \times 10^{-3} * (299460.387 + 2251581.85)}{600}$$

$i = 2.12587$ Amps

Question 3





Given

Mean length of core $l_c = 40\,cm$

$A_c = 12\,cm^2$

$= 12 \times 10^{-4}\,m^2$

$N = 400$

Air gap length $l_g = 0.05\,cm$

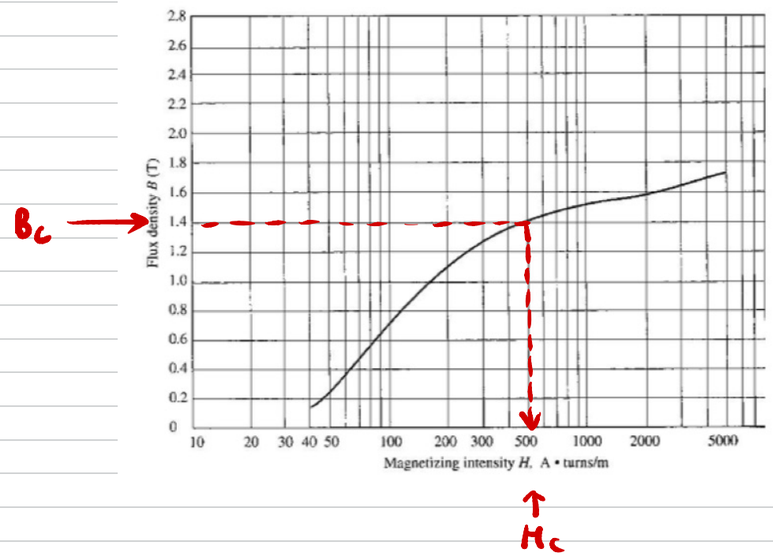
$= 0.05 \times 10^{-2}\,m$

$Ni = H_c l_c + H_g l_g \Rightarrow$ $\boxed{i = \dfrac{H_c l_c + H_g l_g}{N}}$

(a) Air gap flux $\phi_g = 1.68\,mWb$

$\phi_c = \phi_g = 1.68\,mWb$

$B_c = \dfrac{\phi_c}{A_c} = 1.4\,T$



from graph above $B_c = 1.4\,T$

$H_c = 500\,At/m$

with $B_g = B_c$

$$H_g = \frac{B_g}{\mu_0} = 1114084.602$$

$$i = \frac{H_c l_c + H_g l_g}{N}$$

$$= \frac{500 * 40 \times 10^{-2} + 1114084.602 * 0.05 \times 10^{-2}}{400}$$

$$i = 1.8926 \text{ Amps}$$

(b)

$$B_c = \mu_0 \mu_r H_c$$

$$\mu_r = \frac{B_c}{\mu_0 H_c}$$

$$= \frac{1.4}{\mu_0 * 500}$$

$$\mu_r = 2228.169$$

(c) Core Reluctance

$$R_C = \frac{1}{\mu_0 A_C} \frac{l_C}{\mu_r}$$

$$= \frac{1}{\mu_0 * 12\times10^{-4}} \frac{40\times10^{-2}}{2228.619}$$

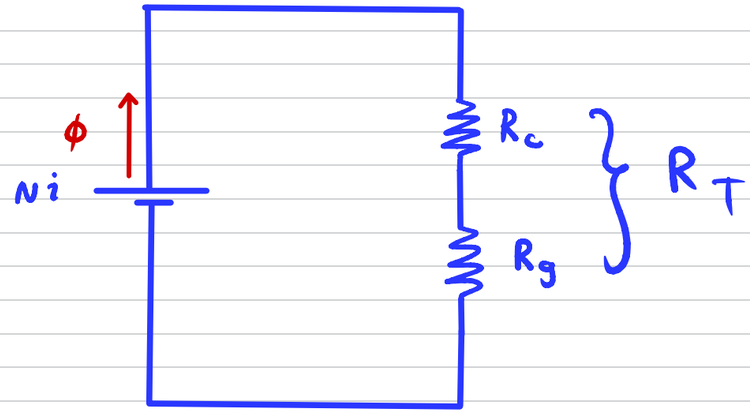
$$= 119047.619 \text{ At/wb}$$

Airgap Reluctance

$$R_g = \frac{1}{\mu_0 A_C} l_g$$

$$= \frac{1}{\mu_0 \times 12\times10^{-4}} \times 0.05\times10^{-2}$$

$$R_g = 331572.7981 \text{ At/wb}$$



Total Reluctance, $R_T = R_C + R_g$

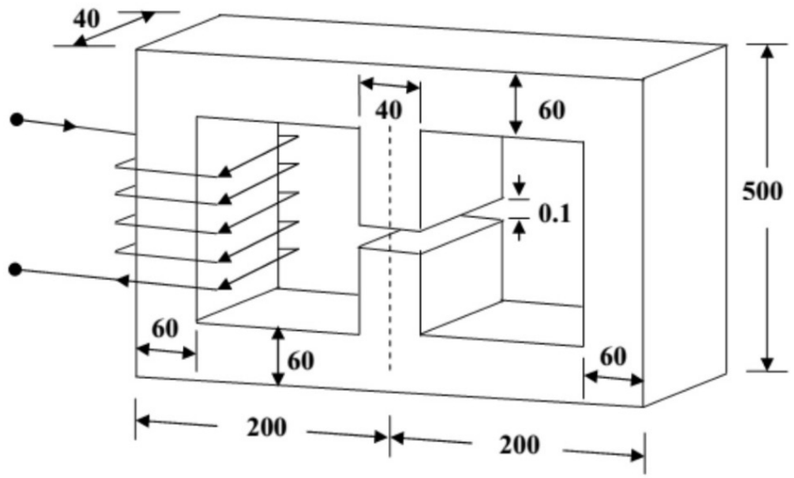
$$= \frac{l_C}{\mu_0 \mu_r A_C} + \frac{l_g}{\mu_r A_C}$$

$$= \frac{1}{\mu_0 A_C}\left(\frac{l_C}{\mu_r} + l_g\right)$$

$$R_T = \frac{1}{\mu_0 \times 12\times10^{-4}} \times \left(\frac{40\times10^{-2}}{2228.619} + 0.05\times10^{-2}\right)$$

$$R_T = 450620.4172 \text{ At/wb}$$

Question 4

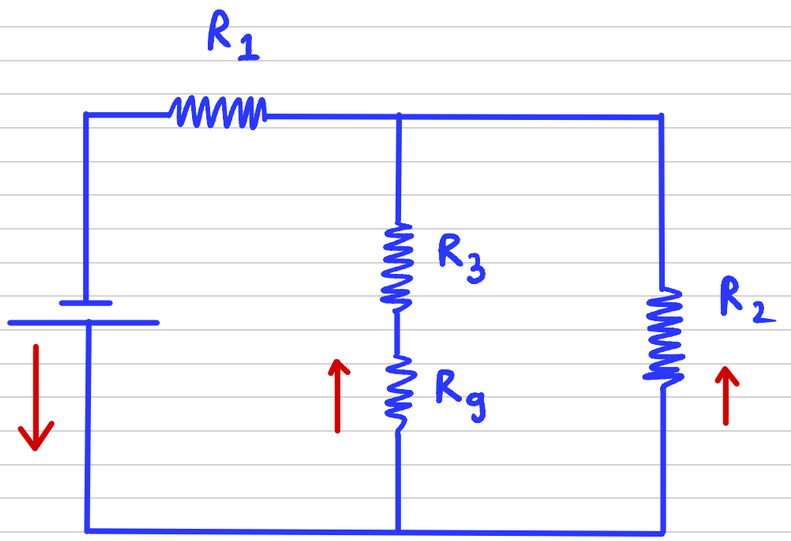


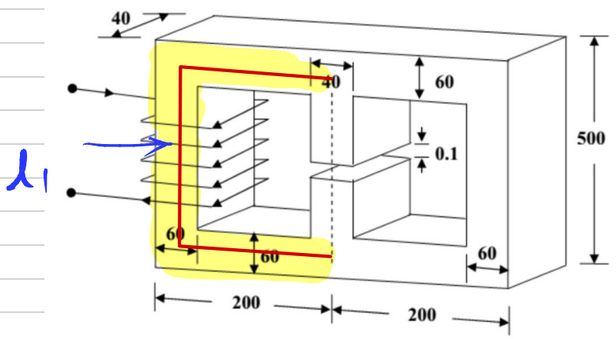
Given
All dimensions are in mm

$N = 200$ turns

$\phi_{air} = \phi_g = 1.28 mwb$

Magnetic circuit



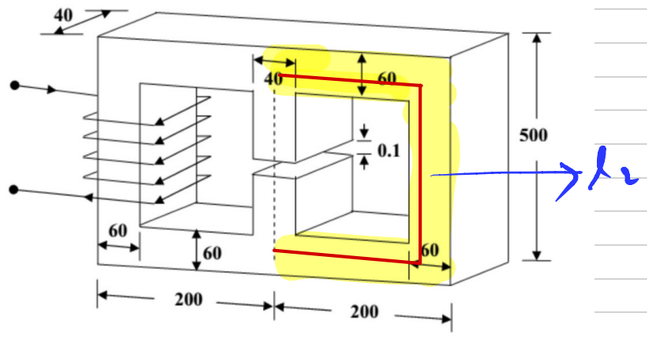


$l_1 = \left(200 - \frac{60}{2}\right) + \left(500 - \frac{60}{2} - \frac{60}{2}\right) + \left(200 - \frac{60}{2}\right)$

$= 780 mm$

$A_1 = (60 \times 10^{-3}) \times (40 \times 10^{-3})$

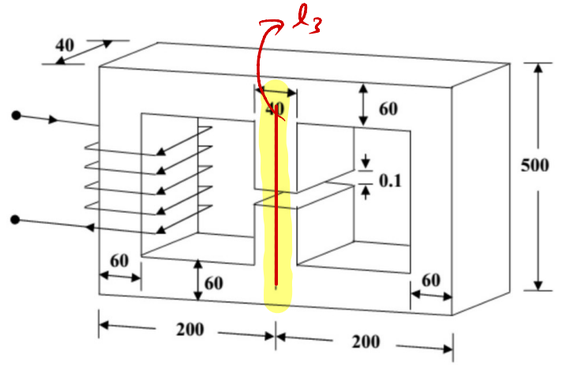
$= 2.4 \times 10^{-3} m^2$



By Observation

$l_2 = l_1 = 780mm$

$A_2 = A_1 = 2.4 \times 10^{-3} m^2$



Given $l_g = 0.1 mm$ (air gap length)

$l_3' = 500 - \frac{60}{2} - l_g - \frac{60}{2}$

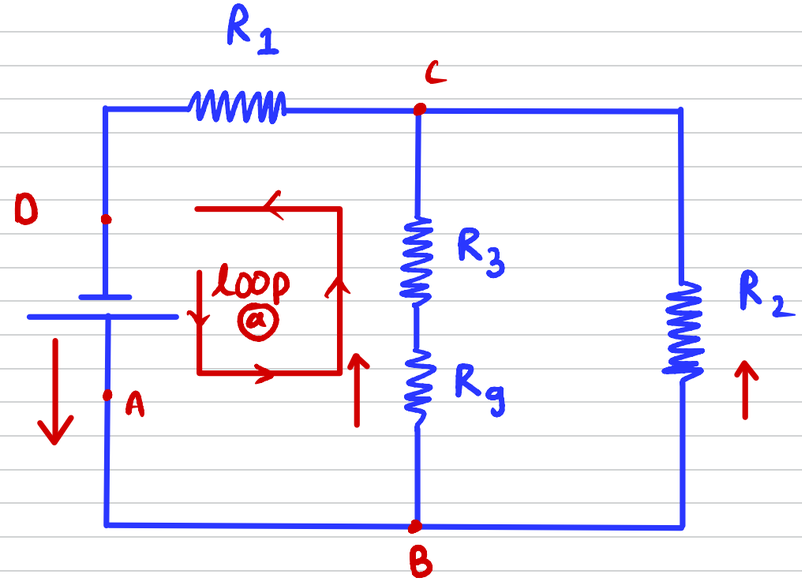
$l_3' = 439.9 mm$

$l_3 = l_3' + l_g$

$= 439.9 + 0.1$

$l_3 = 440 mm$

$A_3 = (40 \times 10^{-3}) \times (40 \times 10^{-3}) = 1.6 \times 10^{-3} m^2$



MMF balance in loop @

$$N \cdot i = MMF_{BC} + MMF_{CD}$$

$$i = \frac{MMF_{BC} + MMF_{CD}}{N}$$

MMF_{BC}

given $\phi_g = 1.28\,mWb$,

$$B_g = \frac{\phi_g}{A_3} = \frac{1.28 \times 10^{-3}}{1.6 \times 10^{-3}}$$

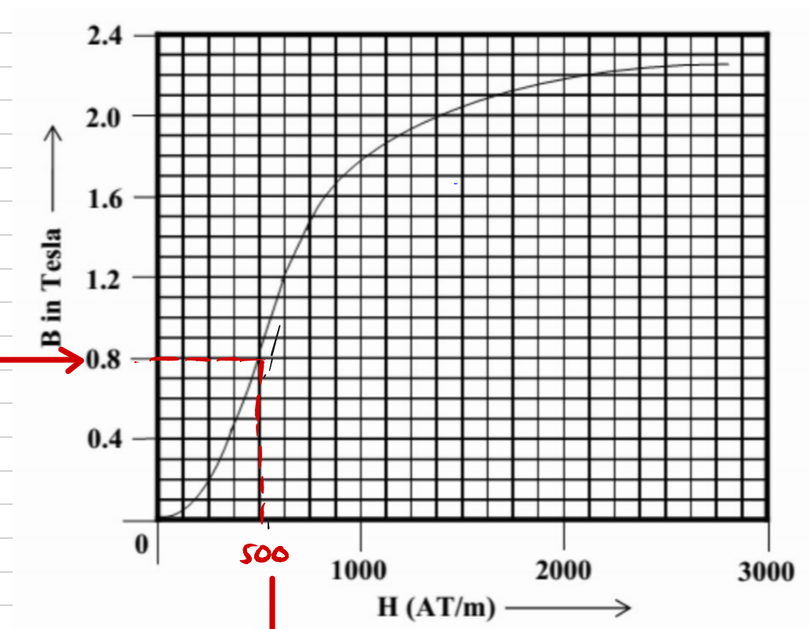
$$B_g = 0.8T$$

$$H_g = \frac{B_g}{\mu_0} = \frac{0.8}{\mu_0}$$

$$H_g = 636619.7724\ At/m$$

$$H_g l_g = 636619.7724 \times 0.1 \times 10^{-3}$$

$$= 63.66198\ At$$



$B_{C3} \simeq B_g = 0.8T$

$H_{C3} \simeq 500\ At/m$

$$H_{C3} l_3' = 500 \times 439.9 \times 10^{-3}$$

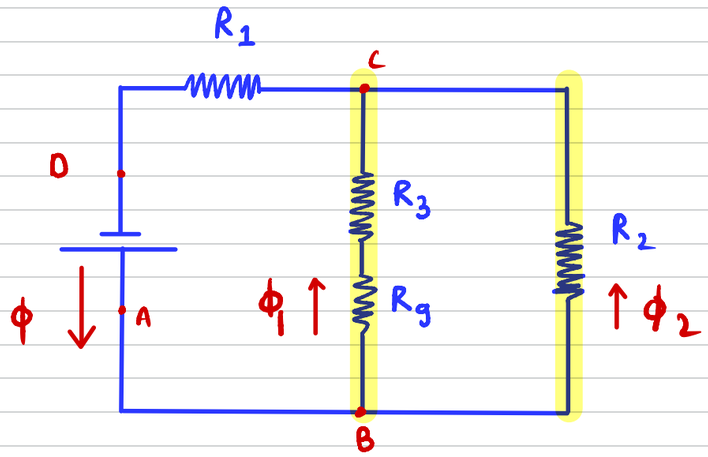
$$= 219.95\ At$$

$$MMF_{BC} = H_g l_g + H_{c3} l_3'$$

$$= 63.66198 + 219.95$$

$$\boxed{MMF_{BC} = 283.61197 \text{ At}}$$

To find MMF_{CD}



$\phi = \phi_1 + \phi_2$

$\phi_1 = 1.28 mWb$

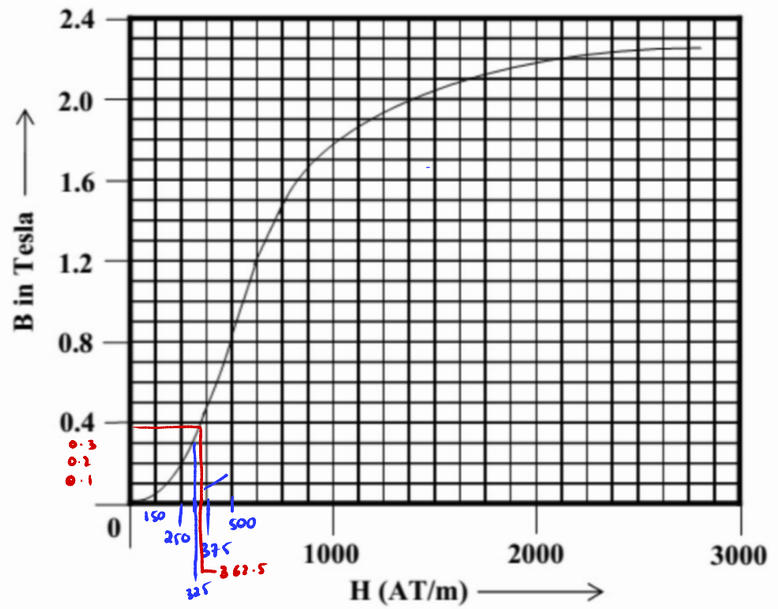
To find ϕ_2

$$MMF_{BC} = H_2 l_2$$

$$\Rightarrow H_2 = \frac{MMF_{BC}}{l_2} = \frac{283.61197}{760 \times 10^{-3}}$$

$$= 363.605 \text{ At/m}$$

From graph $B_2 = 0.39 T$



So,

the flux $\phi_2 = B_2 A_2$

$= 0.39 \times 2.4 \times 10^{-3}$

$= 0.000936$ wb

$\phi_2 = 0.936$ mWb

Now,

$\phi = \phi_1 + \phi_2$

$= 1.28 \times 10^{-3} + 0.936 \times 10^{-36}$

$\phi = 2.216$ mWb

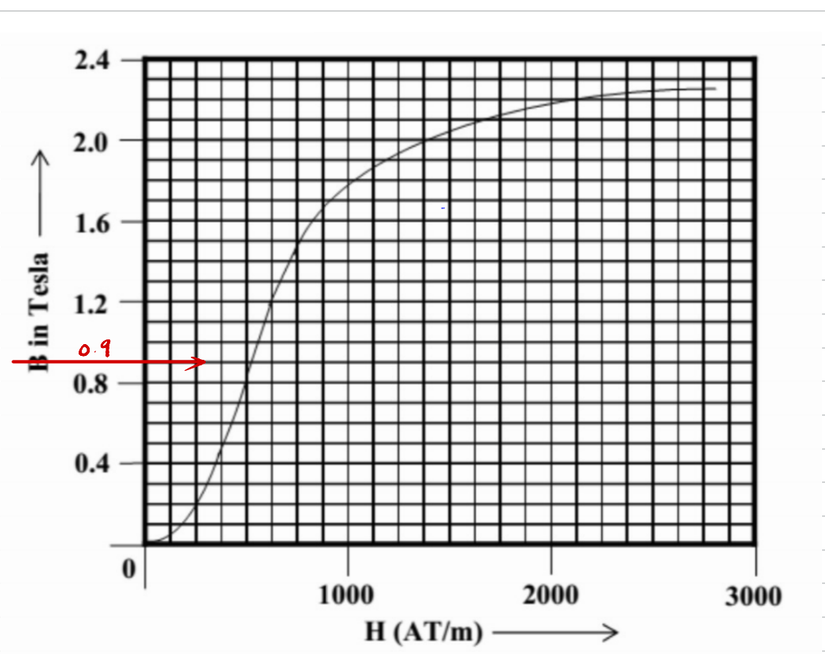
$B_1 = \frac{\phi}{A_1} = \frac{2.216}{2.4 \times 10^{-3}}$

$B_1 = 0.9233$

$H_1 =$ 562.5 AT/m

$MMF_{CD} = H_1 l_1 = 562.5 \times 780 \times 10^{-3}$

$= 438.7$ AT

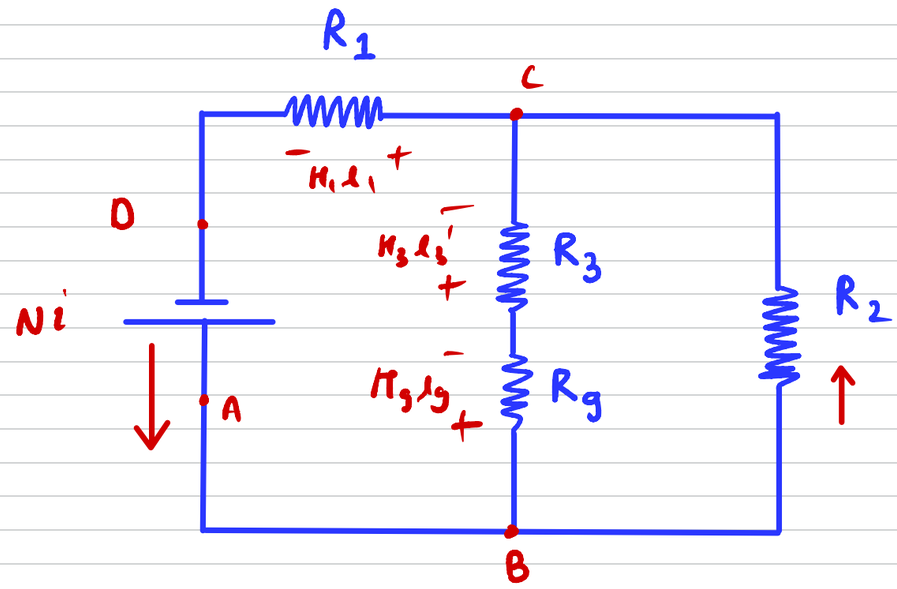


$MMF_{BC} = 283.61197$ At

$MMF_{CD} = 438.7$ At

$$i = \frac{283.61197 + 438.7}{200}$$

$i = 3.611$ Amps



$$i = \frac{MMF_{BC} + MMF_{CD}}{N}$$

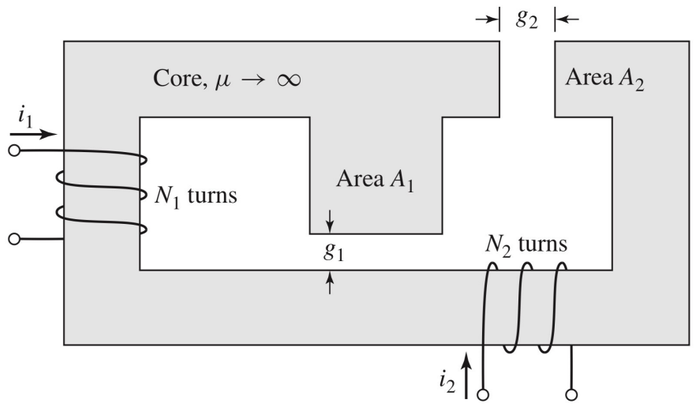
$$i = \frac{H_g l_g + H_2 l_3' + H_1 l_1}{N}$$

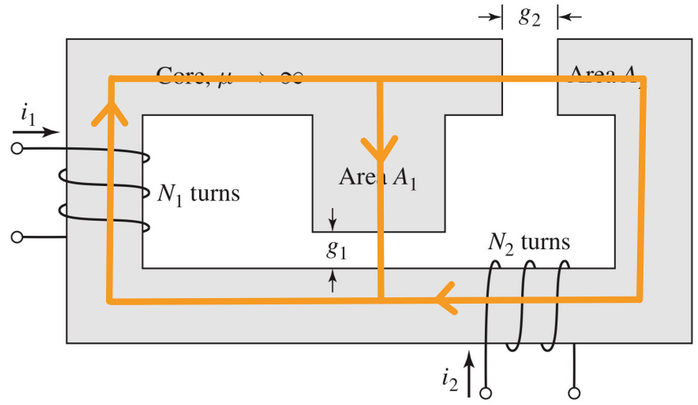
$Ni = H_1 l_1 + H_2 l_2$

$i = \frac{(562.5 + 363.605)\,780 \times 10^3}{200}$

$i = 3.611$ Amps

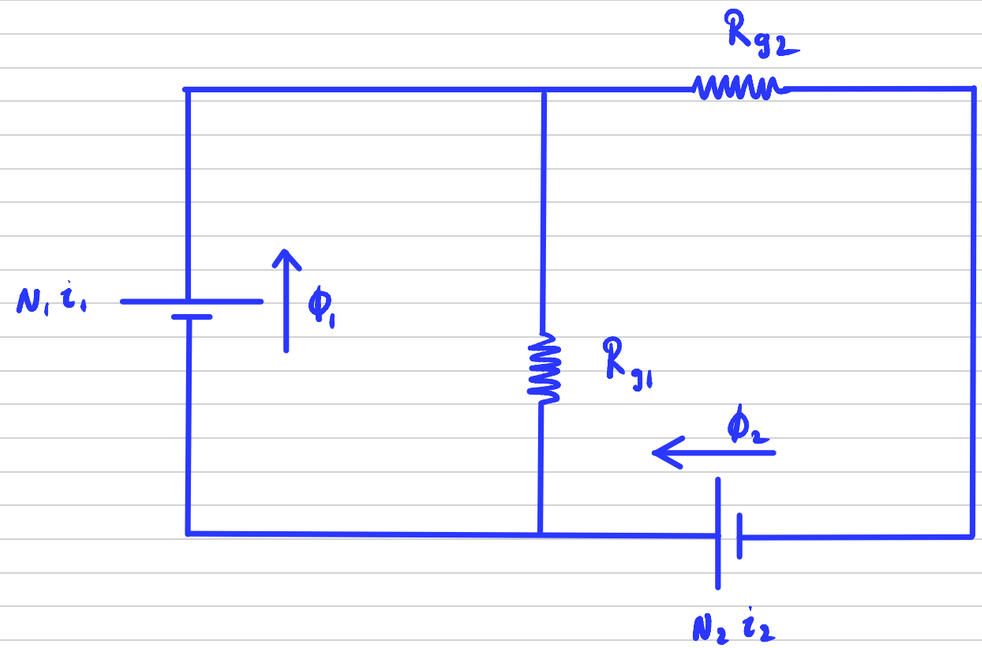
Question 5



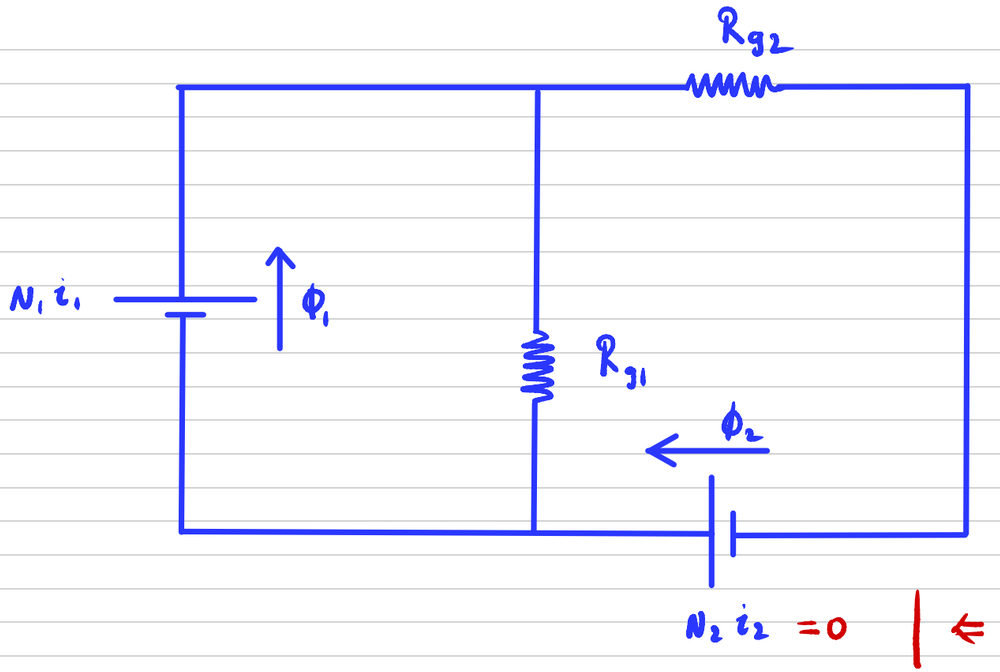


$$R = \frac{l}{\mu A}$$

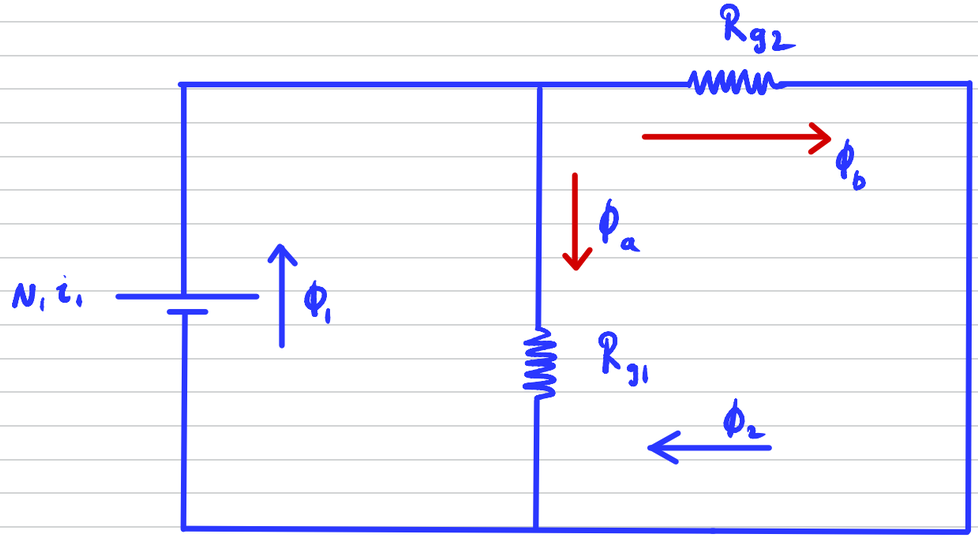
$$\mu = \infty \Rightarrow R = 0$$

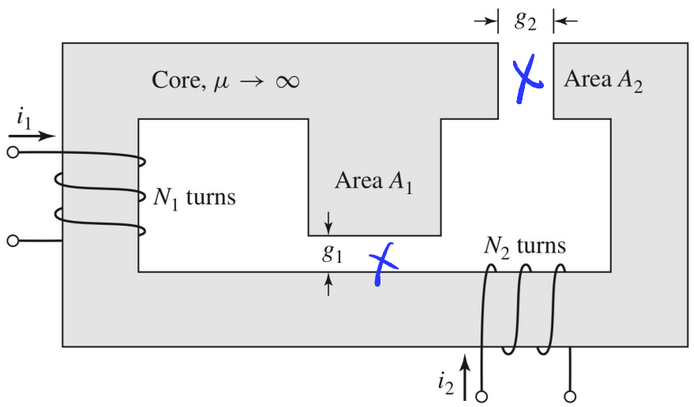


5 (a)



$\Leftarrow i_2 = 0$





$$\phi_a = \frac{N_1 i_1}{R_{g1}} \quad ; \quad \phi_b = \frac{N_1 i_1}{R_{g2}}$$

$$B_{g1} = \frac{\phi_a}{A_1} = \frac{N_1 i_1}{R_{g1}} \times \frac{1}{A_1} = \frac{N_1 i_1}{g_1} \frac{\mu_0 A_1}{A_1} = \frac{\mu_0 N_1}{g_1} i_1$$

$$B_{g2} = \frac{\phi_b}{A_2} = \frac{N_1 i_1}{R_{g2}} = \frac{N_1 i_1}{g_2} \frac{\mu_0 A_2}{A_2} = \frac{\mu_0 N_1}{g_2} i_1$$

$$\boxed{B_{g1} = \frac{\mu_0 N_1}{g_1} i_1}$$

$$\boxed{B_{g2} = \frac{\mu_0 N_1}{g_2} i_1}$$

$$N_1\phi_1 = N_1(\phi_a + \phi_b) = N_1\left(\frac{N_1 i_1}{R_{g1}} + \frac{N_1 i_1}{R_{g2}}\right)$$

$$= N_1\left(\frac{N_1 i_1}{\frac{g_1}{\mu_0 A_1}} + \frac{N_1 i_1}{\frac{g_2}{\mu_0 A_2}}\right)$$

$$\psi_1 = N_1\phi_1 = \mu_0 N_1^2\left(\frac{A_1}{g_1} + \frac{A_2}{g_2}\right) i_1$$

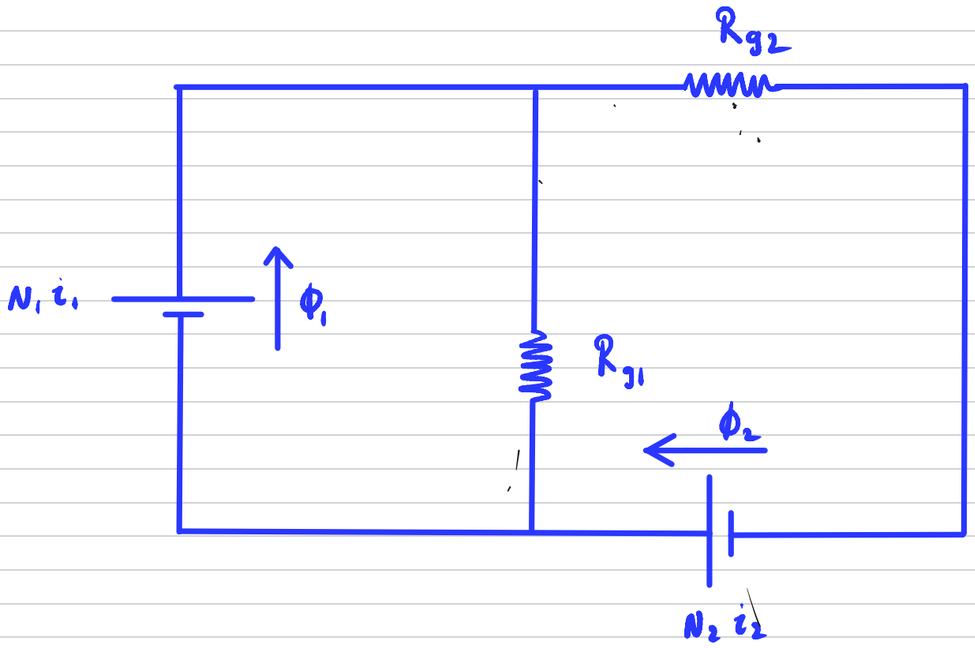
$$N_2\phi_2 = N_2\phi_b = N_2\frac{N_1 i_1}{R_{g2}} = N_1N_2\frac{i_1}{\frac{g_2}{\mu_0 A_2}} = \mu_0 N_1 N_2\frac{A_2}{g_2} i_1$$

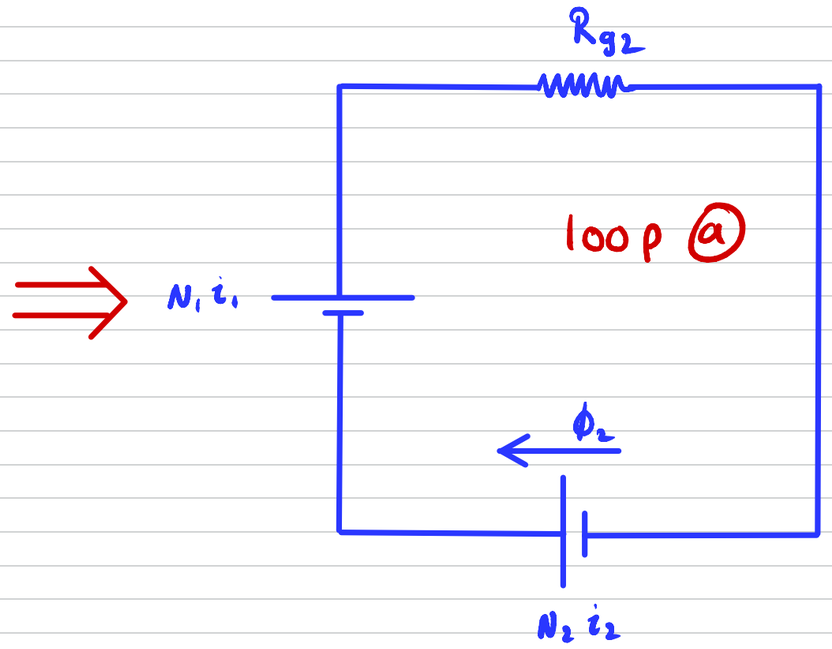
$$\psi_2 = \mu_0 N_1 N_2\frac{A_2}{g_2} i_1$$

$$\psi_1 = \mu_0 N_1^2\left(\frac{A_1}{g_1} + \frac{A_2}{g_2}\right) i_1$$

$$\psi_2 = \mu_0 N_1 N_2\frac{A_2}{g_2} i_1$$

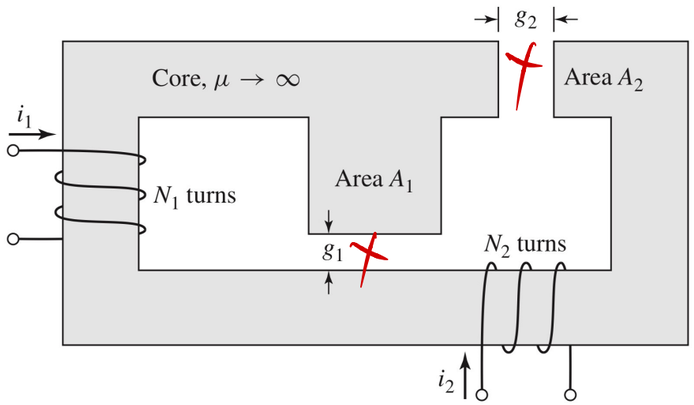
5 (b)





$$N_1 i_1 + N_2 i_2 = \phi_2 R_{g2}$$

$$\phi_2 = \frac{N_1 i_1 + N_2 i_2}{R_{g2}}$$



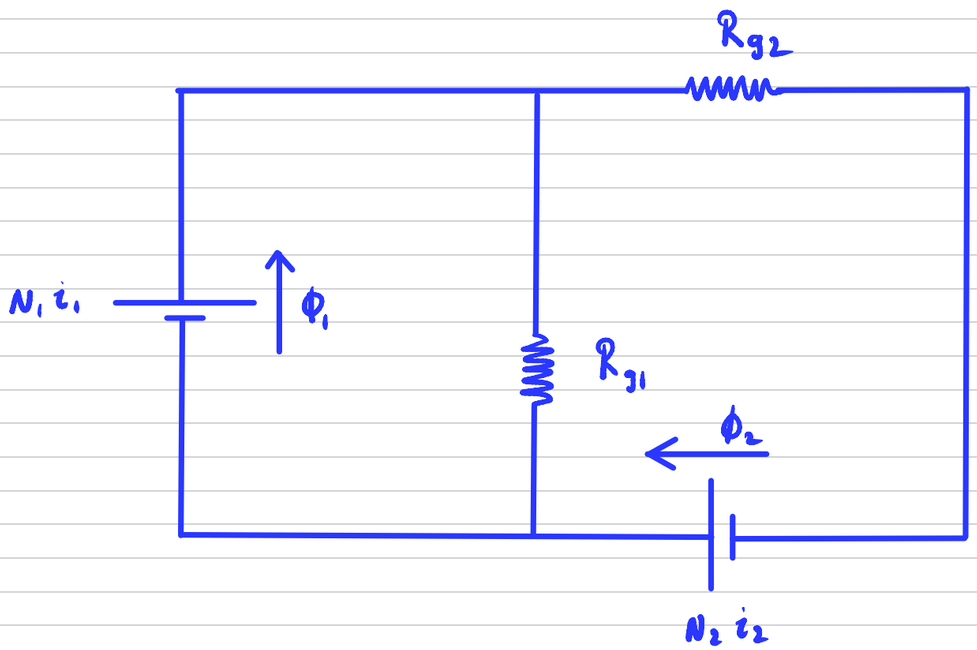
$$\phi_{g1} = \frac{N_1 i_1}{R_{g1}}$$

$$\Rightarrow \quad B_{g1} = \frac{\phi_{g1}}{A_1} = \frac{N_1 i_1}{A_1 R_{g1}} = \frac{N_1 i_1}{A_1} \frac{\mu_0 A_1}{g_1} = \frac{\mu_0 N_1}{g_1} i_1$$

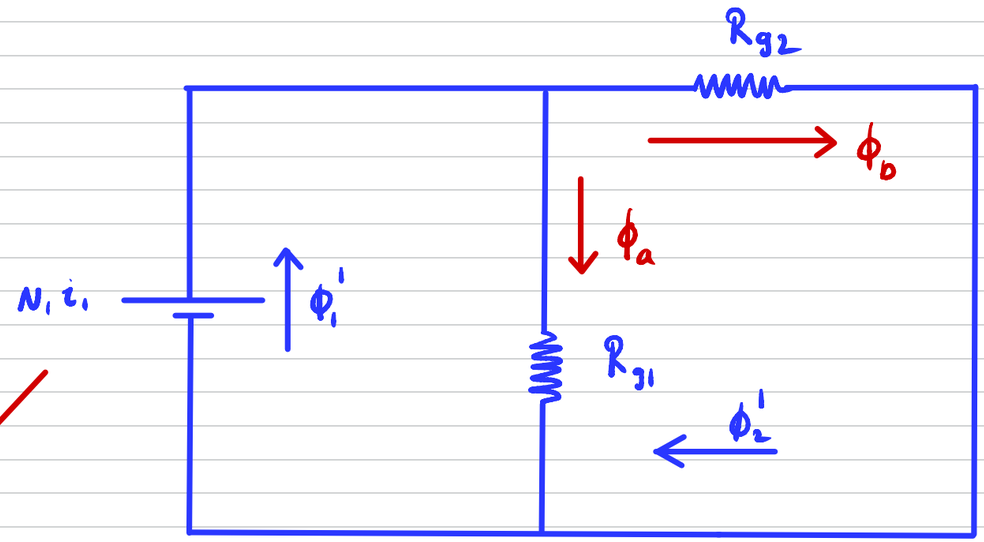
$$B_{g2} = \frac{\phi_2}{A_2} = \left(\frac{N_1 i_1 + N_2 i_2}{R_{g2}} \right) \frac{1}{A_2}$$

$$= \frac{N_1 i_1 + N_2 i_2}{g_2} \times \frac{\mu_0 A_2}{A_2}$$

$$= \frac{\mu_0 N_1}{g_2} i_1 + \frac{\mu_0 N_2}{g_2} i_2$$



$i_2 = 0$

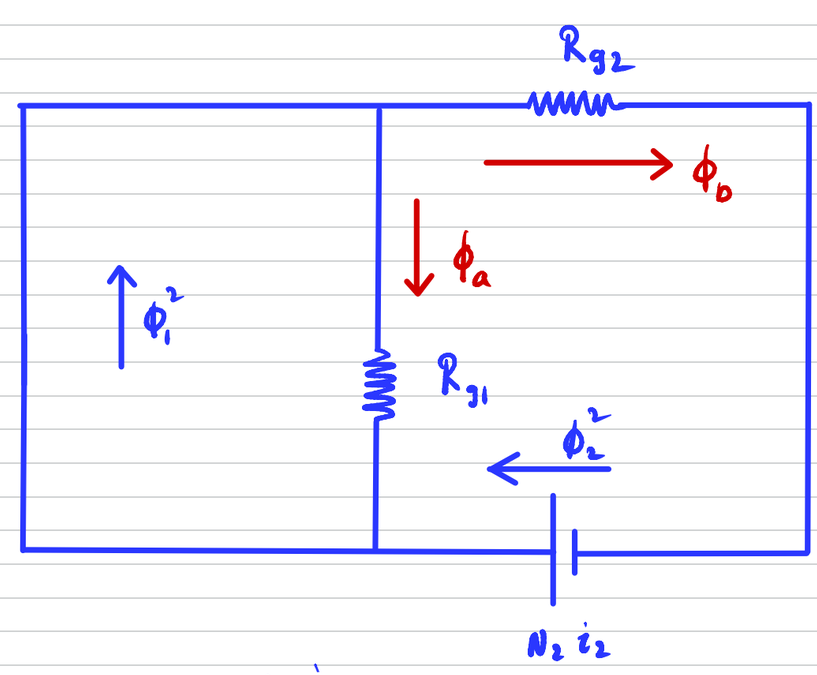


$$\phi_1' = \frac{N_1 i_1}{R_{g1}} + \frac{N_1 i_1}{R_{g2}} \quad ; \quad \phi_2' = \frac{N_1 i_1}{R_{g2}}$$

$$\phi_1 = \phi_1' + \phi_1^2 = \frac{N_1 i_1}{R_{g1}} + \frac{N_1 i_1}{R_{g2}} + \frac{N_2 i_2}{R_{g2}}$$

$$\phi_2 = \phi_2' + \phi_2^2 = \frac{N_1 i_1}{R_{g2}} + \frac{N_2 i_2}{R_{g2}}$$

$i_1 = 0$



$$\phi_1^2 = \frac{N_2 i_2}{R_{g2}} \quad ; \quad \phi_2^2 = \frac{N_2 i_2}{R_{g2}}$$

$$\phi_1 = \frac{N_1 i_1}{R_{g1}} + \frac{N_1 i_1}{R_{g2}} + \frac{N_2 i_2}{R_{g2}}$$

$$\phi_2 = \frac{N_1 i_1}{R_{g2}} + \frac{N_2 i_2}{R_{g2}}$$

$$N_1\phi_1 = \frac{N_1^2 i_1}{R_{g1}} + \frac{N_1^2 i_1}{R_{g2}} + \frac{N_1 N_2 i_2}{R_{g2}}$$

$$= \frac{N_1^2 i_1}{g_1}\mu_0 A_1 + \frac{N_1^2 i_1}{g_2}\mu_0 A_2 + \frac{N_1 N_2 i_2}{g_2}\mu_0 A_2$$

$$= \mu_0 N_1^2\left(\frac{A_1}{g_1} + \frac{A_2}{g_2}\right) + \mu_0 N_1 N_2 \frac{A_2}{g_2} i_2$$

$$N_2\phi_2 = \frac{N_1 N_2 i_1}{R_{g2}} + \frac{N_2^2 i_2}{R_{g2}}$$

$$= \frac{N_1 N_2 i_1}{g_2}\mu_0 A_2 + \frac{N_2^2 i_2}{g_2}\mu_0 A_2$$

$$= \mu_0 N_1 N_2 \frac{A_2}{g_2} i_1 + \mu_0 N_2^2 \frac{A_2}{g_2} i_2$$

$$\psi_1 = \mu_0 N_1^2\left(\frac{A_1}{g_1} + \frac{A_2}{g_2}\right) + \mu_0 N_1 N_2 \frac{A_2}{g_2} i_2$$

$$\psi_2 = \mu_0 N_1 N_2 \frac{A_2}{g_2} i_1 + \mu_0 N_2^2 \frac{A_2}{g_2} i_2$$

Question 6

Given data,

	Stator	Rotor	Airgap
Mean length	$l_s = 50\ cm$	$l_r = 5\ cm$	$l_g = 0.05\ cm$
Cross sectional area	$A_{cs} = 12\ cm^2$	$A_{cr} = 12 cm^2$	$A_g = 14 cm^2$

μ_r (iron) = 2000

N = 200 turns

i = 1 Amp

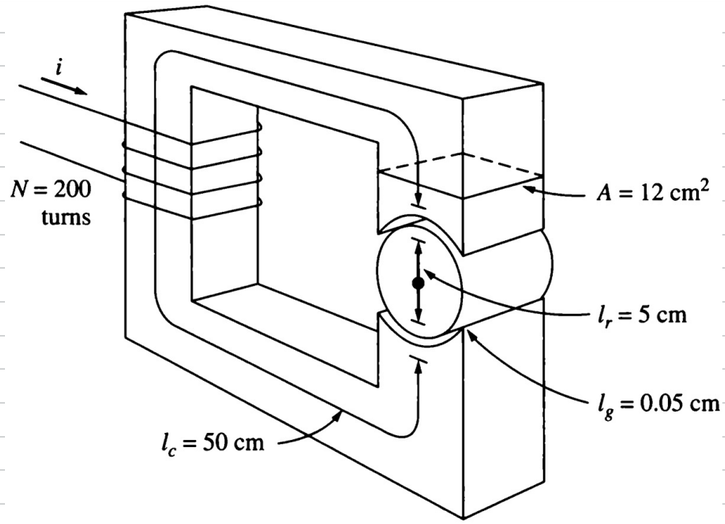
Air gap flux ϕ_g = ?

$$R_s = \frac{l_s}{\mu_0 * \mu_r \times A_{cs}} = \frac{50 \times 10^{-2}}{\mu_0 \times 2000 \times 12 \times 10^{-4}}$$

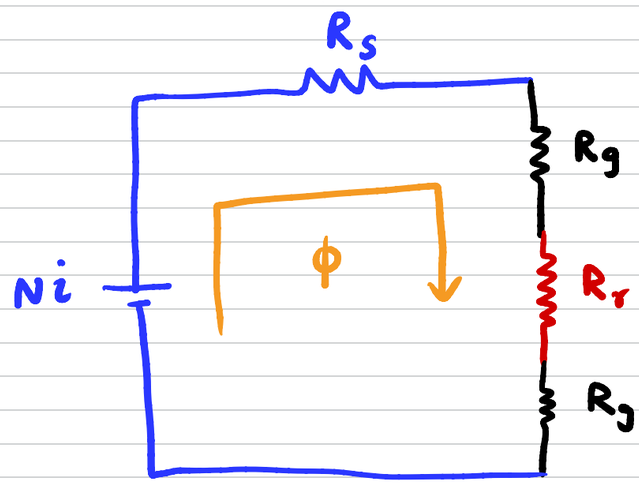
$$= 165786.3991 \ At/wb$$

$$R_r = \frac{l_r}{\mu_0 \times \mu_r \times A_{cr}} = \frac{5 \times 10^{-2}}{\mu_0 \times 2000 \times 12 \times 10^{-4}}$$

$$= 16578.63991 \ At/wb$$



Magnetic Circuit



$$R_g = \frac{l_g}{\mu_0 \times \mu_{rg} \times A_g} = \frac{0.05 \times 10^{-2}}{\mu_0 \times 1 \times 14 \times 10^{-4}}$$

$$= 284205.2555 \ At/wb$$

$$\phi_c = \frac{Ni}{R_s + R_r + 2 * R_g}$$

$$= \frac{200 \times 1}{165786.3991 + 16578.63991 + 284205.2555 \times 2}$$

$$\phi_c = 0.000266$$

$$\phi_c = 0.2664 \text{ mWb}$$

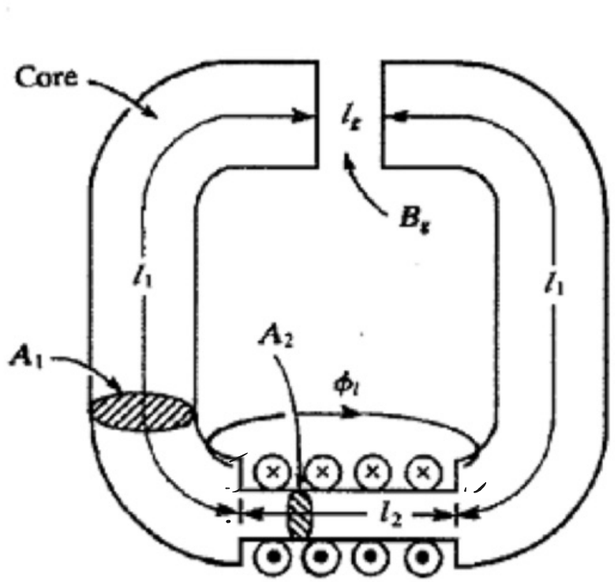
$$\phi_g = \phi_c$$

Air gap flux density

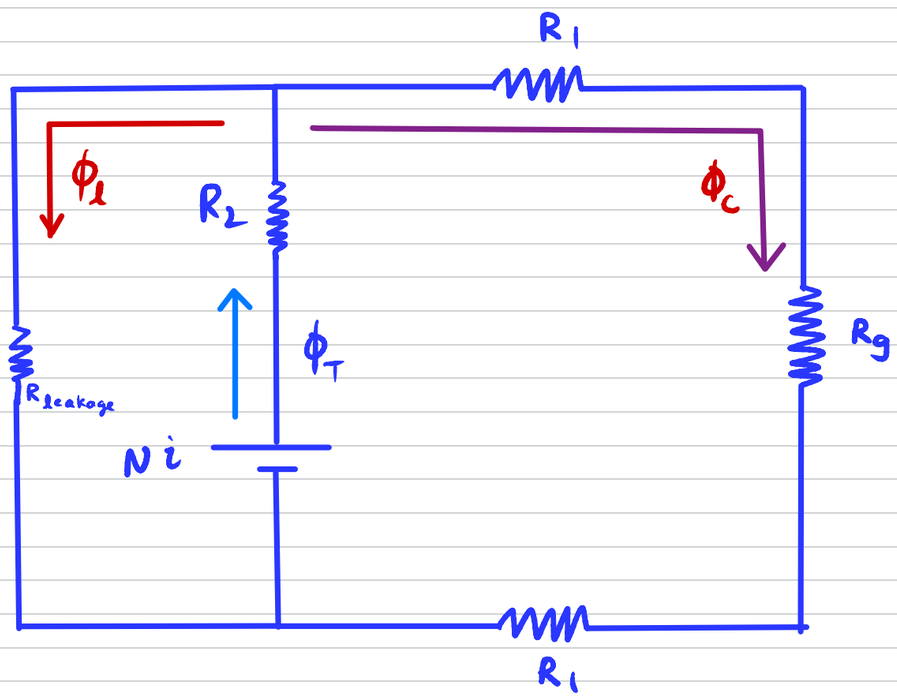
$$B_g = \frac{\phi_g}{A_g} = \frac{0.2664 \times 10^{-3}}{14 \times 10^{-4}}$$

$$B_g = 0.1903 \text{ T}$$

Question 7



Magnetic circuit



Given

$l_1 = +l_2 = 40cm$ | $l_1 = 40cm$

$l_2 = \frac{40}{4} = 10cm$

$A_1 = 2A_2 = 10cm^2$ | $A_1 = 10 \times 10^{-4} m^2$

$A_2 = \frac{10}{2} \times 10^{-4} m^2$

$= 5 \times 10^{-4} m^2$

$l_g = 2mm = 2 \times 10^{-3} m$

Leakage flux, $\phi_l = 0.01 mwb$

Air gap flux density, $B_g = 0.6T$

$N = 400$ turns

$B_g = 0.6\ T \Rightarrow \phi_g = B_g * A_g$

$A_g = A_l = 10 \times 10^{-4}\ m^2$ $= 0.6 * 10 * 10^{-4}$

$= 6 \times 10^{-4}$

$\phi_c = \phi_g = 6 \times 10^{-4}\ wb$

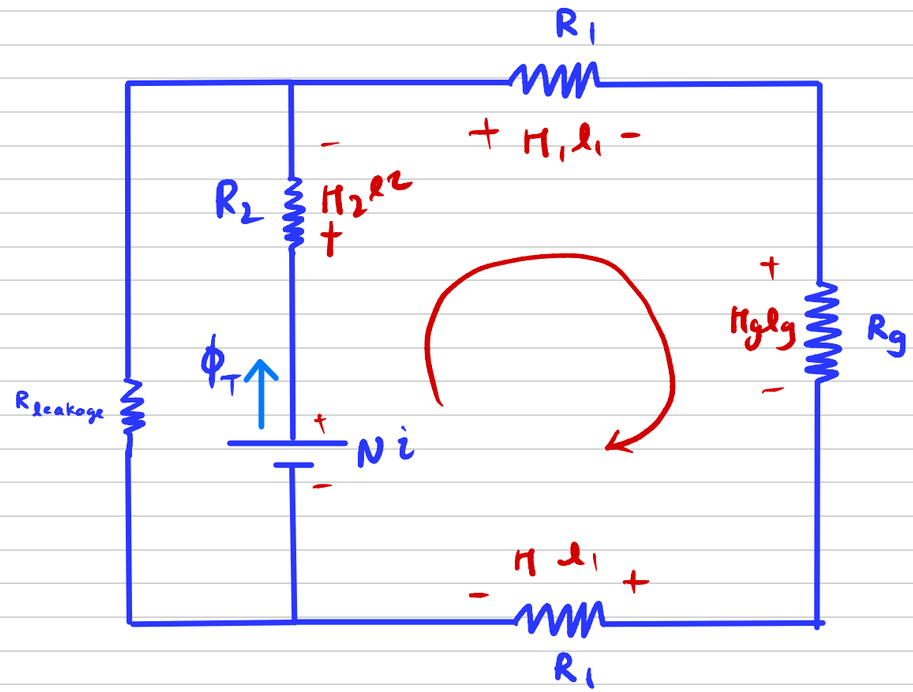
$\phi_T = \phi_c + \phi_l = 6 \times 10^{-4} + 0.01 \times 10^{-3}$

$\boxed{\phi_T = 6.1 \times 10^{-4}\ wb}$

MMF balance

$Ni = \phi_l + \phi_c (R_{l_1} * 2 + R_{l_2} + R_g)$

$\boxed{\mu_1, \mu_2 = ?}$



MMF Balance

$Ni = \sum Hl$

$Ni = H_1 l_1 + H_2 l_2 + H_g l_g + H_1 l_1$

Calculating H_g:

We need to maintain air gap flux density,

$$B_g = 0.6\ T \quad \text{(given)}$$

now $$H_g = \frac{B_g}{\mu_0} \quad (\because \mu_r\ (air) = 1)$$

$$\boxed{H_g = +77464.8293\ A/m}$$

$\phi R = B * A * \frac{l}{\mu_0 \mu_r A}$ $\quad B = \mu H$

$= \frac{B}{\mu_0 \mu_r} l$

$\phi R = Hl$

$Ni = \sum H_i l_i$

$Ni = H_1 l_1 + H_2 l_2 + H_g l_g + H_3 l_3$

$\Sigma \phi R_i = \sum H_i l_i \Rightarrow \Sigma H_i l_i$

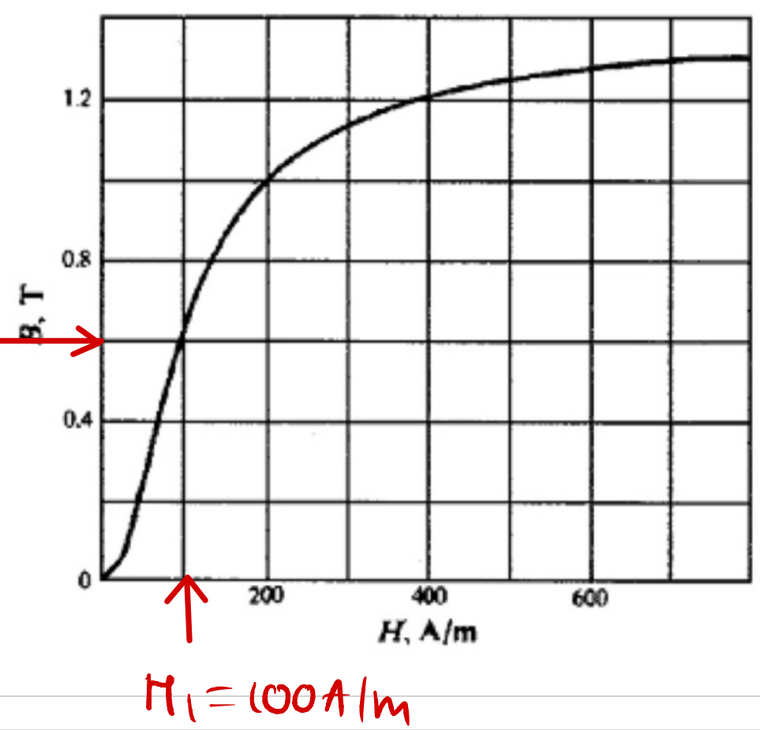
To find H_1:

we know $\phi_c = \phi_g$ flow through core of length l

$$B_g = B_1 = 0.6\ T$$

from graph

$$\boxed{H_1 \simeq 100\ At/m}$$



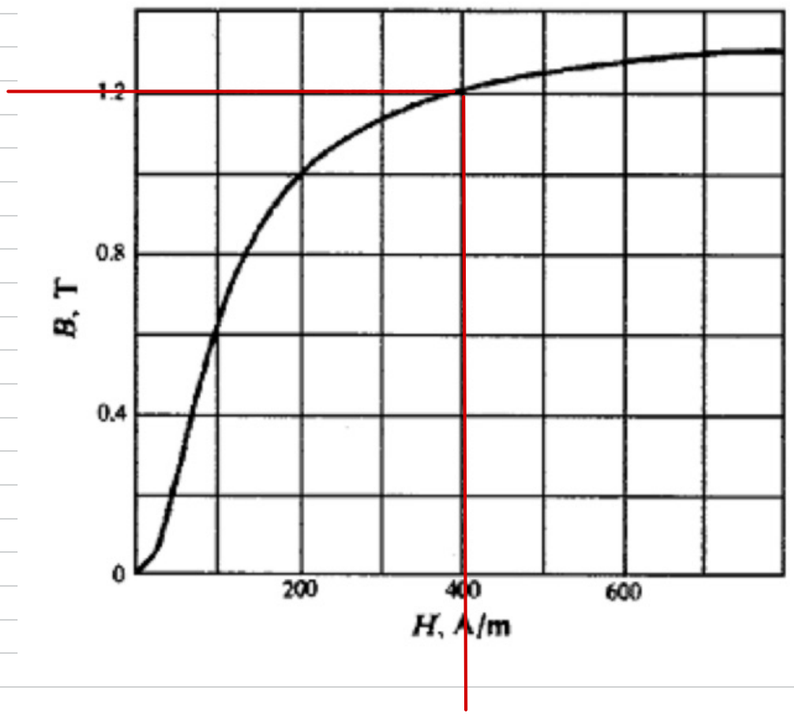
To find H_2

$$B_2 = \frac{\phi_T}{A_2} = \frac{6.1 \times 10^{-4}}{5 \times 10^{-4}}$$

$$= 1.22\ T \simeq 1.2\ T$$

from graph

$$H_2 \simeq 400\ T \quad \text{(approx.)}$$



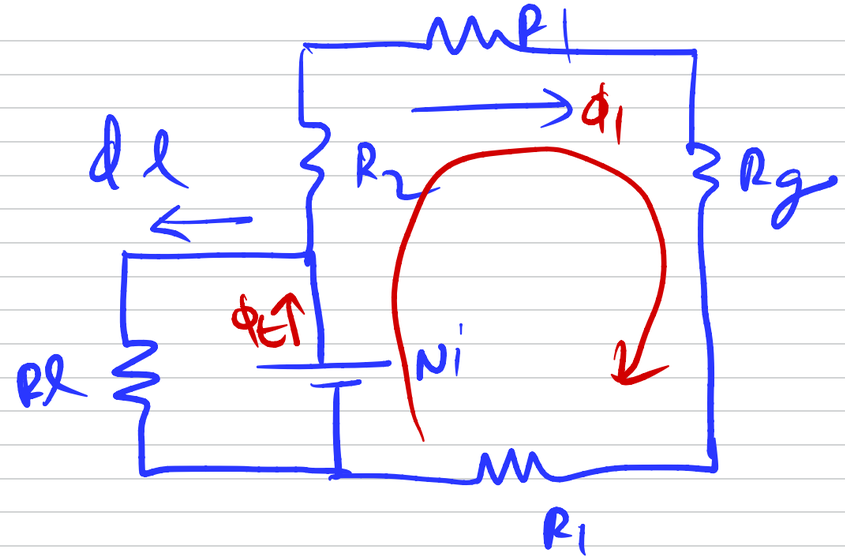
$$\therefore \quad \sum_i H_i l_i = Ni$$

$$H_1 l_1 + H_g l_g + H_2 l_2 + H_1 l_1 = Ni$$

$$i = \frac{2 H_1 l_1 + H_2 l_2 + H_g l_g}{N}$$

$$= \frac{2 \times 100 \times 40 \times 10^{-2} + 400 \times 10 \times 10^{-2} + \frac{477464.829 \times}{2 \times 10^{-3}}}{400}$$

$$\boxed{i = 2.6873\ \text{Amps}}$$



$\phi_1 = 6 \times 10^{-4}$

$\phi_l = 0.01 \times 10^{-3}$

$\phi_t = \phi_1 + \phi_l = 6.1 \times 10^{-4}$ Wb

$Ni = 2 H_1 l_1 + H_2 l_2 + H_g l_g$ $\quad H_1 = H_2$

$$i = \frac{1}{N}\left(H_1 (2l_1 + l_2) + \frac{B_g}{\mu_0} l_g \right)$$

$$= \frac{1}{400}\left(100 \left(2 \times 40 \times 10^{-2} + 10 \times 10^{-2}\right) + \frac{0.6}{\mu_0} \times 2 \times 10^{-3} \right)$$

$$= 2.6123 \text{ Amps}$$